

# Lesson 5: Limits of Transcendental Functions

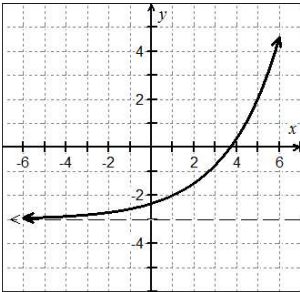
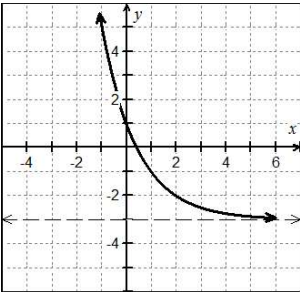
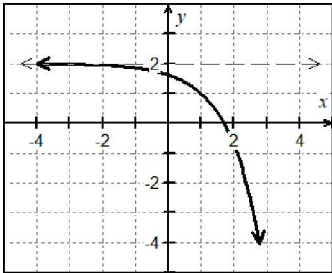
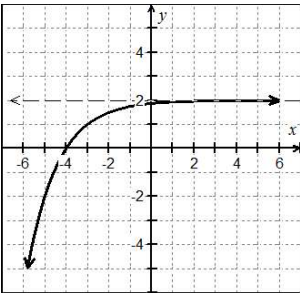
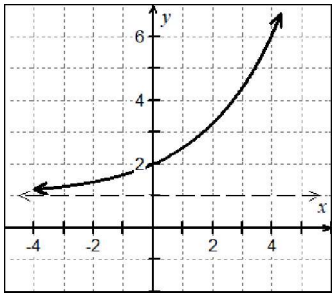
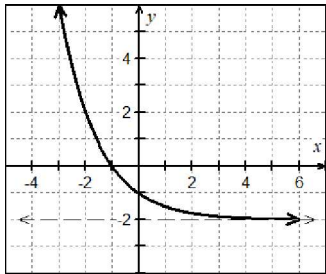
If you research the definition of a **transcendental function** you will find this explanation on Wolfram's Math World:

*A function which is not an algebraic function. In other words, a function which "transcends," i.e., cannot be expressed in terms of, algebra. Examples of transcendental functions include the exponential function, the trigonometric functions, and the inverse functions of both.*

## Topic 1.9: Connecting Multiple Representations of Limits

### Analyzing Limits of Exponential Functions

**EX #1:** Recall that exponential equations are written in the form  $y = ab^{(x-h)} + k$ . You will need to find limits of exponential functions without the aid of a graph or calculator in this course. Do you remember the rules for transformations of exponential functions? Evaluate the limits using the graphs and look for patterns.

| Exponential Growth                                                                                                                                                                                                                                                        | Exponential Decay                                                                                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>A.</b> <math>f(x) = \left(\frac{3}{2}\right)^{x-1} - 3</math></p> <p><math>\lim_{x \rightarrow -\infty} f(x) =</math></p> <p><math>\lim_{x \rightarrow \infty} f(x) =</math></p>  | <p><b>D.</b> <math>f(x) = \left(\frac{1}{2}\right)^{x-2} - 3</math></p> <p><math>\lim_{x \rightarrow -\infty} f(x) =</math></p> <p><math>\lim_{x \rightarrow \infty} f(x) =</math></p>    |
| <p><b>B.</b> <math>f(x) = -e^{x-1} + 2</math></p> <p><math>\lim_{x \rightarrow -\infty} f(x) =</math></p> <p><math>\lim_{x \rightarrow \infty} f(x) =</math></p>                       | <p><b>E.</b> <math>f(x) = -\left(\frac{1}{2}\right)^{x+3} + 2</math></p> <p><math>\lim_{x \rightarrow -\infty} f(x) =</math></p> <p><math>\lim_{x \rightarrow \infty} f(x) =</math></p>  |
| <p><b>C.</b> <math>f(x) = \left(\frac{2}{3}\right)^{-x} + 1</math></p> <p><math>\lim_{x \rightarrow -\infty} f(x) =</math></p> <p><math>\lim_{x \rightarrow \infty} f(x) =</math></p>  | <p><b>F.</b> <math>f(x) = 2^{-x} - 2</math></p> <p><math>\lim_{x \rightarrow -\infty} f(x) =</math></p> <p><math>\lim_{x \rightarrow \infty} f(x) =</math></p>                           |

**EX #2: WHAT JUST HAPPENED?** Did you see that? Basically, the end-behavior of any exponential function tends toward three places.

| CASE #1: | CASE #2: | CASE #3: |
|----------|----------|----------|
|          |          |          |

In PreCalculus, we learned the following rules without considering the behavior for negative  $a$ -values.

| $y = ab^{(x-h)} + k$                                  |                                                      |
|-------------------------------------------------------|------------------------------------------------------|
| $a > 0$ and $b > 1$                                   | $a > 0$ and $0 < b < 1$                              |
| $f$ is increasing<br>$b$ is exponential growth factor | $f$ is decreasing<br>$b$ is exponential decay factor |

**Growth and decay are not synonymous with increasing and decreasing.**

Consider the behavior of the function in relation to the horizontal asymptote. Notice whether the function is moving away from or toward the horizontal asymptote.

Also, what effect does the negative coefficient on  $x$  do to the function behavior?

**EX #3:** Now, use the equations from example #1 to complete the table below:

| Graph of $f(x)$                              | Base        | Increase or Decrease | $\lim_{x \rightarrow -\infty} f(x)$ | $\lim_{x \rightarrow \infty} f(x)$ |
|----------------------------------------------|-------------|----------------------|-------------------------------------|------------------------------------|
| <b>GROWTH</b>                                | $b > 1$     |                      |                                     |                                    |
| $f(x) = \left(\frac{3}{2}\right)^{x-1} - 3$  |             |                      |                                     |                                    |
| $f(x) = -e^{x-1} + 2$                        |             |                      |                                     |                                    |
| $f(x) = \left(\frac{2}{3}\right)^{-x} + 1$   |             |                      |                                     |                                    |
| <b>DECAY</b>                                 | $0 < b < 1$ |                      |                                     |                                    |
| $f(x) = \left(\frac{1}{2}\right)^{x-2} - 3$  |             |                      |                                     |                                    |
| $f(x) = -\left(\frac{1}{2}\right)^{x+3} + 2$ |             |                      |                                     |                                    |
| $f(x) = 2^{-x} - 2$                          |             |                      |                                     |                                    |

**EX #4:** You got this! Find the limits of each of the following exponential functions.

|                                                |                                                                                 |
|------------------------------------------------|---------------------------------------------------------------------------------|
| A. $\lim_{x \rightarrow -\infty} -(0.3)^x + 2$ | B. $\lim_{x \rightarrow -\infty} e^{-x+1} - 3$                                  |
| C. $\lim_{x \rightarrow \infty} -3^{-x+1} - 3$ | D. $\lim_{x \rightarrow \infty} -\left(\frac{1}{3}\right)^{-x} + 1$             |
| E. $\lim_{x \rightarrow 2} (e^{x-3} - 1)$      | F. $\lim_{x \rightarrow -3} \left[ \left(\frac{1}{2}\right)^{-x-1} + 5 \right]$ |

## Analyzing Limits of Trigonometric Functions

Most of the work you have done with limits to this point have dealt with polynomial or rational functions. When confronted with trigonometric functions, you will find throughout the course, that there are several methods to use. Let's begin with the basics in this section.

### Topic 1.7: Selecting Procedures for Determining Limits

**EX#5:** Use direct substitution to find each limit.

|                                                                          |                                                              |
|--------------------------------------------------------------------------|--------------------------------------------------------------|
| A. $\lim_{\theta \rightarrow \frac{\pi}{3}} \frac{\cos 2\theta}{\theta}$ | B. $\lim_{\theta \rightarrow \frac{\pi}{2}} 3 \sin^2 \theta$ |
|--------------------------------------------------------------------------|--------------------------------------------------------------|

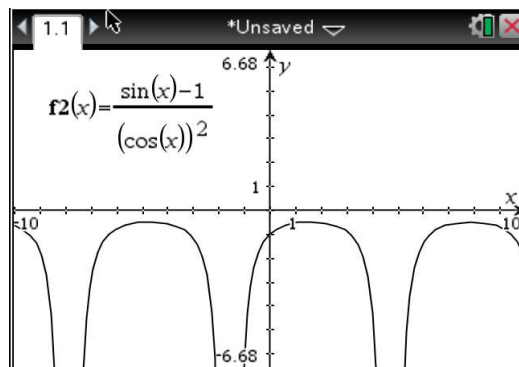
Just like some polynomial functions where a function value is not defined, yet a limit will exist...the same will occur with trigonometric functions.

**EX #6: Evaluate the limit by direct substitution.**  $\lim_{\theta \rightarrow \frac{\pi}{2}} \frac{\sin \theta - 1}{\cos^2 \theta}$

## A few thoughts to consider:

If you algebraically determine that a function is undefined at a value, does that mean the limit does not exist? Can you look at the graph? ***How do you analyze the limit when no calculator or graph is permitted?***

We need to use some trig identities and rewrite the function. Try that here:



### Topic 1.6: Determining Limits Using Algebraic Manipulation

**EX #7: You Got This!** Use your new skills to evaluate each limit below by first rewriting the function and using identities.

A.  $\lim_{\theta \rightarrow \frac{\pi}{2}} 2 \cos \theta \cot \theta$

B.  $\lim_{\theta \rightarrow \frac{\pi}{4}} \frac{\sin \theta \cos \theta}{\tan \theta}$

C.  $\lim_{\theta \rightarrow \pi} \frac{\sin \theta \csc \theta}{3\theta}$

**HAVE YOU DISCOVERED A PATTERN IN ALL OF THIS YET? If you ALWAYS try direct substitution first, three things will occur!**

Later in the course you will learn a smooth technique known as L'Hospital's Rule. It will be a quicker method for evaluating many functions. For now, you might like to memorize these special rules, as well.

### SPECIAL TRIGONOMETRIC LIMITS

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$$

$$\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta} = 0$$

**EX #8:** Use a graphing calculator and the table feature to evaluate the following trigonometric limits.

|                                                                   |      |       |   |      |     |
|-------------------------------------------------------------------|------|-------|---|------|-----|
| A. $\lim_{\theta \rightarrow 0} \frac{\sin 2\theta}{\theta}$      | -0.1 | -0.01 | 0 | 0.01 | 0.1 |
|                                                                   |      |       |   |      |     |
| B. $\lim_{\theta \rightarrow 0} \frac{\cos 3\theta - 1}{\theta}$  | -0.1 | -0.01 | 0 | 0.01 | 0.1 |
|                                                                   |      |       |   |      |     |
| C. $\lim_{\theta \rightarrow 0} \frac{\sin 3\theta}{4\theta}$     | -0.1 | -0.01 | 0 | 0.01 | 0.1 |
|                                                                   |      |       |   |      |     |
| D. $\lim_{\theta \rightarrow 0} \frac{1 - \cos 3\theta}{2\theta}$ | -0.1 | -0.01 | 0 | 0.01 | 0.1 |
|                                                                   |      |       |   |      |     |

**EX#9: Some algebraic tricks with properties.**

A.  $\lim_{\theta \rightarrow 0} \frac{3 \sin 5\theta}{2\theta}$

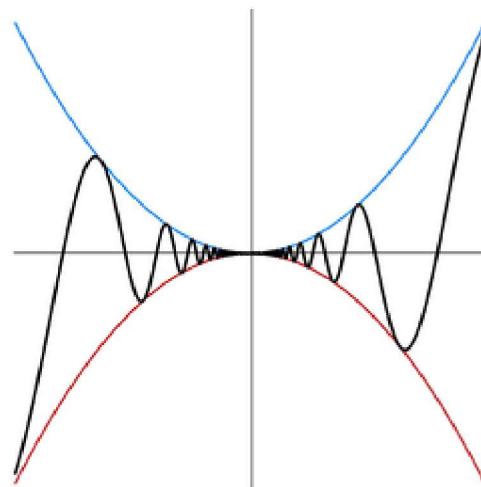
B.  $\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta + \sin 3\theta}{\theta}$

## Topic 1.8: Determining Limits Using the Squeeze Theorem

**EX #10: Evaluate the limit using the squeeze theorem.**

$$\lim_{x \rightarrow 0} \left[ (x^2) \sin \left( \frac{1}{x} \right) \right]$$

**A.** Explain why you cannot use the Product Limit Law.



**B.** Consider the range of the sine function  $-1 \leq \sin x \leq 1$ . We can conclude that  $-1 \leq \sin \left( \frac{1}{x} \right) \leq 1$ . Multiply through by  $x^2$  to use the Squeeze Theorem.

**EX #11: Find the limit.**  $\lim_{x \rightarrow 0} \left[ (x^2) \cos \left( \frac{1}{x^2} \right) \right]$